Vojenský výzkumný ústav, s. p.

VERIFICATION OF DETECTAIR'S HYDROfogg SYSTEM DECONTAMINATION EFFICACY OF SURFACES CONTAMINATED WITH VX AGENT

(contract no. 302324/027; order no. 202382)

By:
Marek ANDRLE, Ph.D.
Karel MAZANEC, Ph.D.

Brno 2023

1 TABLE OF CONTENTS

1 TABLE OF CONTENTS 2
2 Experimental section 3
2.1 Instruments and equipment 3
2.2 Chemical warfare agents 3
2.3 Chemicals 3
2.4 Experimental procedure 3
3 Results and disussion 7
4 Conclusion 8
APPENDIX A. CWA-Certificate of analysis A-1
B. STANAG 4653 National Authority Certification B-1

Tesla DETECTAIR s. r. o. company contracted Military Research Institute, state enterprise - VVÚ, s. p. to verify decontamination efficacy of HYDROfogg system. The experimental work was performed on November 1st, 2023.

2 EXPERIMENTAL SECTION

2.1 INSTRUMENTS AND EQUIPMENT

(a) gas chromatograph 8890, *Agilent Technologies*, with dual FPD detector;
(b) analytical balances, *KERN*, ABJ220-4NM;
(c) electronic pipette, *Eppendorf Research*, objem 0,5 ÷ 10 μl;
(d) pipettes with adjustable volume, *Eppendorf Research*, objem 10, 20, 100, 200, 1 000, 5 000 μl;
(e) air thermostat, type FTD100, *ISCO*;
(f) Dispensette dispenser, *BRAND*, dosing volume 5 ÷ 50 ml;
(g) Adsorbent material: sheets (5 × 5 cm) Al-foil with a layer of fine-grained silica gels for TLC (DC Alufolien Kieselgel 60W), *MERCK*;
(h) vortex VS 15 O, *Lauda*;
(i) contamination stamp with needles 0,8 mm, *VVÚ, s. p.*;
(j) weight 500 g;
(k) glass panels 5 × 5 cm;
(l) panels coated with paint system for interior surfaces 5 × 5 cm, *COLORLAK a.s.* (S2008/0600, S2320/0600, U2056/1000);
(m) stopwatch, *Roth*.

2.2 CHEMICAL WARFARE AGENTS

(a) VX agent, *S*-{2-[Di(propan-2-yl)amino]ethyl} O-ethyl methylphosphonothioate. Purity 70,13 %.

2.3 CHEMICALS

(a) isopropanol, p.a. *Merci*;
(b) heptane, p.a., *Penta*;
(c) acetone, p.a., *Penta.*

2.4 EXPERIMENTAL PROCEDURE

The surface panels (Figure 1) were contaminated with droplets of VX agent using a "contamination stamp" (Figure 2) with an initial density of contamination of 0.2 $g.m^{-2}$, which is the prescribed contamination for internal surfaces of military equipment. The samples were immediately placed in a 79 x 30 x 32 cm (75.8 liter) test chamber. Subsequently, a dry fog of hydrogen peroxide produced by the HYDROfogg device was introduced in to the test chamber (Figure 3). One contaminated sample was not exposed to fog (positive control - PC). The operating parameters of the applied fog were set directly by the customer. The fog exposure time was 30 minutes.

After the fog exposure time, silica gel samples were placed on the panels, weighed with weights (Figure 4) and placed in a thermostat at a temperature of 30 °C (± 0.5 °C) for 15 minutes. Then, the surface panels and silica gel samples were transferred to extraction vessels, overlaid with 20 ml of heptane-acetone mixture (ratio 9:1) and extracted on a shaker for 60 minutes. The concentration of the VX agent residuals in extracts was determined by a gas chromatography.

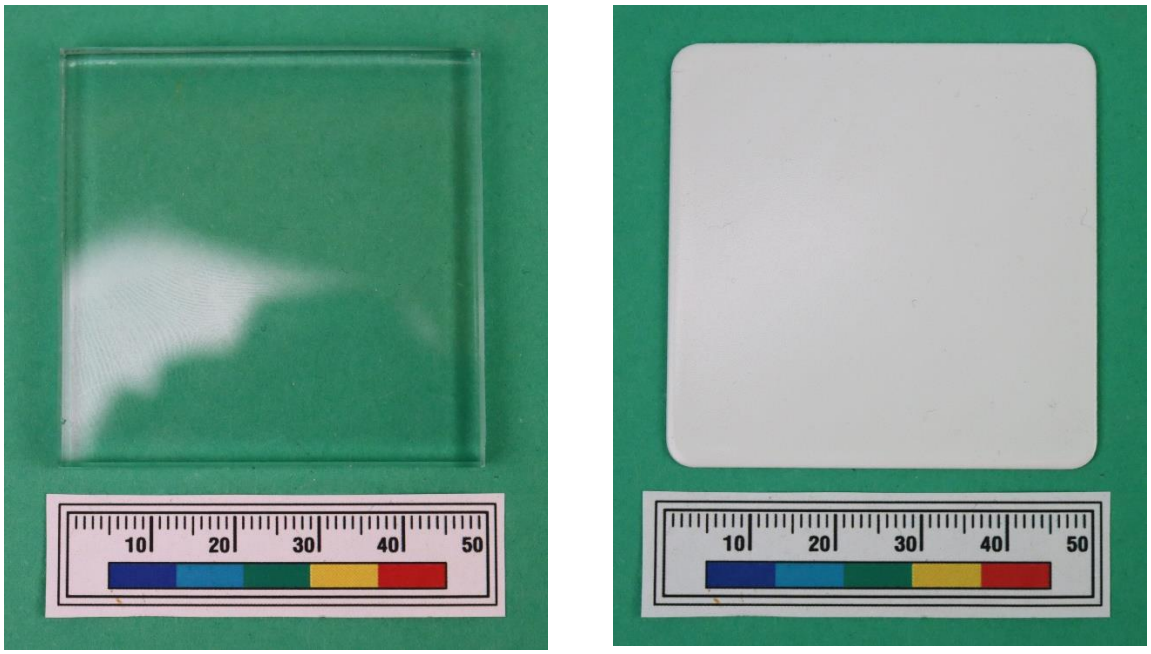


Figure 1 Test panels: glass (left), interior coating system (right)

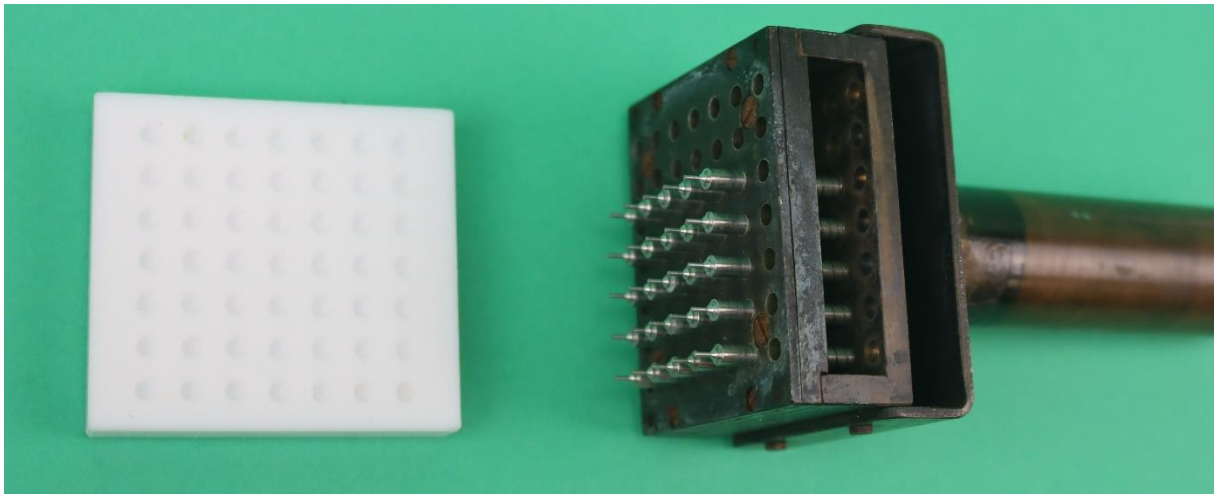


Figure 2 Contamination stamp with reservoir for CWAs

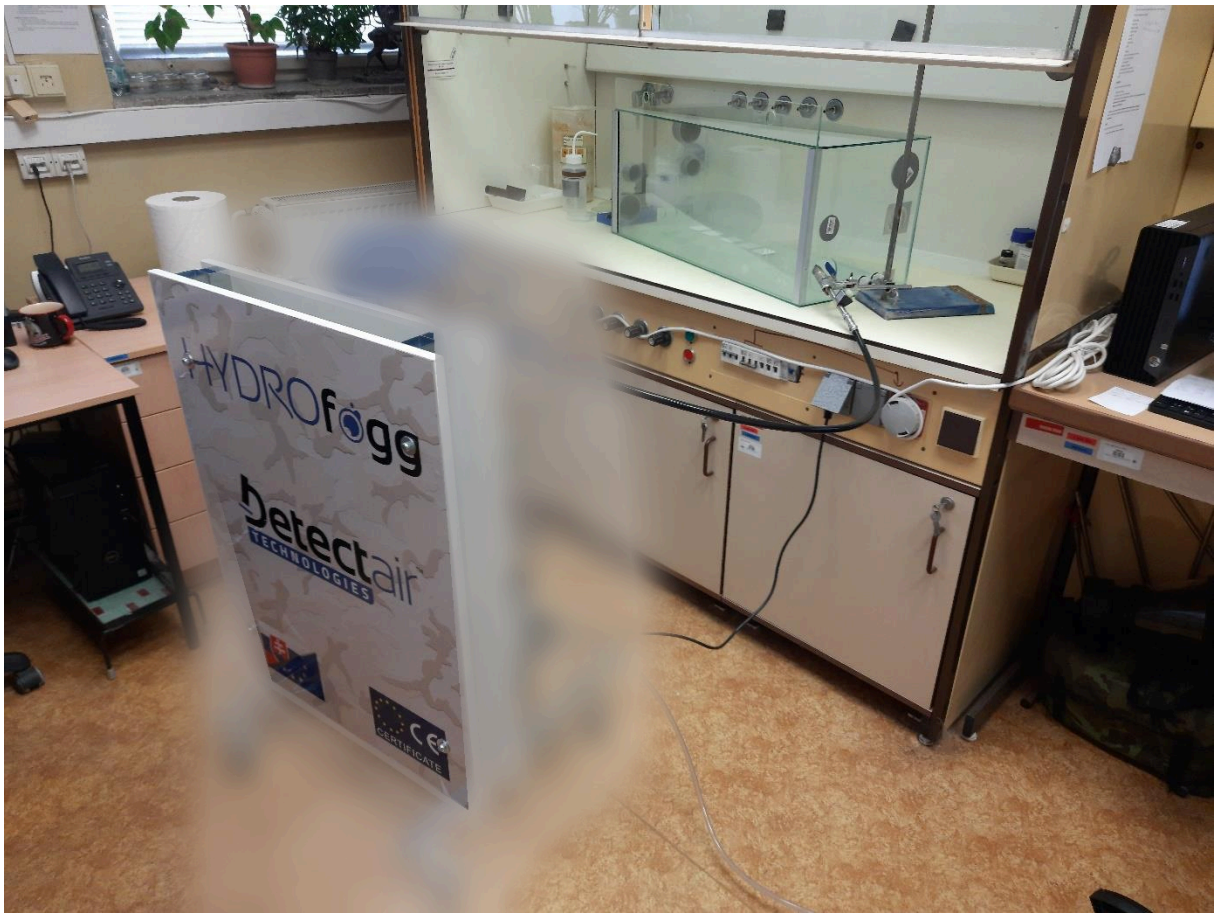


Figure 3 HYDROfogg system connected to test chamber



Figure 4 Coupon with silica gel adsorbent and 500 g weight

Agilent 8890 series Gas Chromatograph with Autosampler, HP7693 Injector was used for the quantification of CWA residuals in extracts. The GC was equipped with a HP-5 column with guard column (Agilent, or equivalent, 5% Phenyl Methyl Siloxane) fused silica capillary column 30 m long x 0.32 mm internal diameter x 0.25 µm film thickness.

Table 1: Gas chromatograph operating conditions

Inlet		
Temperature 250 °C	Septum purge flow 3 ml/min	
Inlet mode: Splitless	Purge flow to split vent 60 ml/min	
Detector		
Temperature 250 °C	Emission block 125 °C	
Air flowrate 60 ml/min		
H_2 flowrate 60 ml/min		
N_2 flowrate 60 ml/min		
Column		
HP-5, length 30 m, inner diameter 0,32 mm, layer thickness 0,25 µm		
Flowrate 3,5 ml/min	Pressure 16 psi	
Pec		
1	2	3
Initial temperature (°C) and hold time (min)	Temperature gradient (°C/min) and final temperature (°C)	Temperature gradient (°C/min) and final temperature (°C)
90 / 1	22 / 210	50 / 310

3 RESULTS AND DISUSSION

To meet the requirement for successful decontamination, it is necessary to reduce the value of residual contamination on the surface below the X value specified in the alliance document. If the value of criterion X is exceeded, the value of criterion Y (judge), the desorption rate of the respective CWAs from the tested material, must also be determined.

If the value of criterion Y set in the alliance document is not exceeded, the safety of persons who may come into direct (unprotected) contact with the surfaces of decontaminated materials for a short time - for a maximum of 15 minutes - is ensured. If the limit of criterion Y is met, the tested decontamination agent (mixture, system etc.) can be considered as sufficiently effective.

The alliance document setting the values of criteria X and Y is classified, therefore the results will only state whether the measured surface contamination after the decontamination by fog meets or does not meet these criteria.

The following tables summarize the obtained data.

Table 2 Measured concentrations of VX agent residuals in extracts from glass panels

	Replicate	c_{GC} ($\mu g.ml^{-1}$)	VX amount (µg / sample)	Contamin. density ($\mu g.cm^{-2}$)	Average contamin. density ($\mu g.cm^{-2}$)	Fulfillment of demands
	PC	23.835	476.700	19.068	19.068	-
X	1	0.176	3.520	0.141	0.060	YES
	2	0.085	1.700	0.068		
	3	0.045	0.900	0.036		
	4	0.050	1.000	0.040		
	5	0.021	0.420	0.017		
Y	1	0.136	2.720	0.109	0.039	YES
	2	0.045	0.900	0.036		
	3	0.020	0.400	0.016		
	4	0.024	0.480	0.019		
	5	0.017	0.340	0.014		

Table 3 Measured concentrations of VX agent residuals in extracts from coated panels

	Replicate	c_{GC} ($\mu g.ml^{-1}$)	VX amount (μg / vzorek)	Contamin. density ($\mu g.cm^{-2}$)	Average contamin. density ($\mu g.cm^{-2}$)	Fulfillment of demands
	PC	24.909	498.180	19.927	19.927	-
X	1	0.103	2.060	0.082	0.095	YES
	2	0.187	3.740	0.150		
	3	0.182	3.640	0.146		
	4	0.077	1.540	0.062		
	5	0.045	0.900	0.036		
Y	1	0.032	0.640	0.026	0.027	YES
	2	0.030	0.600	0.024		
	3	0.051	1.020	0.041		
	4	0.027	0.540	0.022		
	5	0.029	0.580	0.023		

The results presented in tables 2 and 3 confirm that the action of the HYDROfogg device will reduce surface contamination below the permissible limit to the maximum VX agent contamination density for both tested surfaces (glass and interior coating system). Surfaces treated in this way meet the requirements for long-term skin contact hazard (X) and vapour desorption rate (Y). The determined concentration of VX agent residuals was 0.06 $\mu g.cm^{-2}$ on the glass panel with 0.039 $\mu g.cm^{-2}$ transferred into the adsorption material. The residual contamination on the internal coating system was 0.095 $\mu g.cm^{-2}$ with 0.027 $\mu g.cm^{-2}$ desorbed in 15 min.

4 CONCLUSION

The decontamination efficacy of the DETECTAIR's HYDROfogg system was verified on the surfaces contaminated with VX agent. Two types of surfaces, glass and internal coating system, were tested with initial contamination level of 0.2 $g.m^{-2}$. The fog exposure time was 30 minutes. The experiments confirmed the sufficient decontamination efficacy of the HYDROfogg device against the VX agent. The requirements for maximum surface contamination density were met for both surfaces used in the test.

Vojenský výzkumný ústav, s. p.

Certificate of Analysis

VX – V Agent

IUPAC Name:
N-[2-[ethoxy(methyl)phosphoryl]sulfanylethyl]-N-propan-2-ylpropan-2-amine

CAS: 50782-69-9

Molecular Formula: $C_{11}H_{26}NO_2PS$

Molecular Weight: 267.368 $g.mol^{-1}$

Density: 1.0083 $g.cm^{-3}$ at 25 °C

Melting Point: -51 °C

Boiling Point: 298 °C

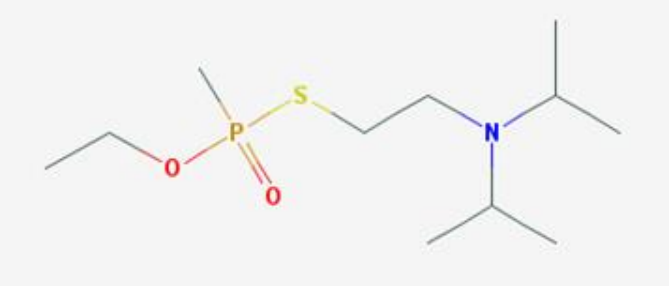
LD_{50}: 8 $mg.kg^{-1}$

LCt_{50}: 35-45 $mg.min.m^{-3}$

ICt_{50}: 5 $mg.min.m^{-3}$

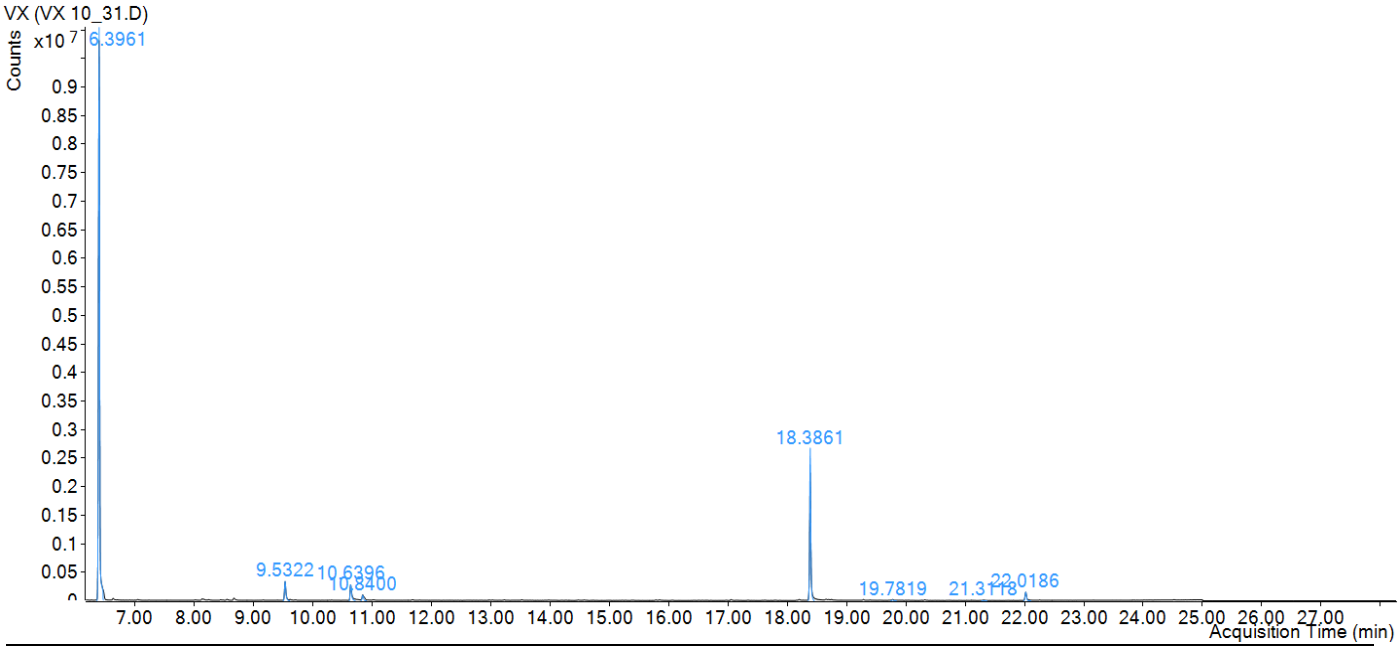
Color:
Colorless to straw colored liquid, similar in appearance to motor oil

Structure:



Volatility:
1.6 $mg.m^{-3}$

Odor:
Odorless



Component RT	Compound Name	Match Factor	CAS#	*Area %*	Component Area
6,3961	Trifluoroacetamide, 2TMS derivative	97,7	21149-38-2	-	14758492,3
9,5322	2-Propanamine, N-(2-chloroethyl)-N-(1-methylethyl)-	97,3	96-79-7	11,33	592818,3
10,6396	2-(Diisopropylamino)ethanethiol	90,7	5842-07-09	9,60	502345,9
10,84	O-Ethyl O-trimethylsilyl methylphosphonothionate	92,4	1000385-77-1	3,15	164669,8
18,3861	***VX***	***93,3***	***50782-69-9***	***70,13***	***3669707,3***
19,7819	Bis(2-diisopropylaminoethyl) sulfide	81,9	110501-56-9	0,58	30502,3
21,3118	3-Hexanone, 2,5-dimethyl-	81,9	1888-57-9	0,19	10072,8
22,0186	Bis(diisopropylaminoethyl)disulfide	96,6	65332-44-7	5,01	262325,7
23,3638	Phosphonothioic acid, methyl-, O-[2-[bis(1-methylethyl)amino]ethyl]-, O-ethyl ester	79	71840-25-0	11,33	14758492,3

Vojenský výzkumný ústav, s. p.
Veslařská 230
637 00 Brno
tel.: +420 543 562 101
fax: +420 543 562 100
WWW: www.vvubrno.cz
IČ: 29372259 DIC: CZ29372259
Companies Register kept at the Regional Court
Brno, mark A25718

Purity: **70.13 %**

Batch no.: VX 10_31

Date of Analysis: October 31, 2023

Vojenský výzkumný ústav, s.p.
Veslařská 230, 637 00 Brno
DIČ: CZ29372259, IČ: 29372259

Marek Andrle, Ph.D.
Department of the Decontamination

CZECH REPUBLIC
Ministry of Defence

Director of Division of Capabilities Development and Planning
issues to

VVÚ, s. p. Brno

CERTIFICATION

to act as National Authority in the area of

Combined operational characteristics, technical specifications, test procedures and evaluation criteria for chemical, biological, radiological and nuclear decontamination equipment (AEP-58) (STANAG 4653)

This Certification is valid within the validity period of STANAG 4653.

Prague: 12 October 2015

BG Pavel ADAM